

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			Temperature	1			1		1
	(b)	(i)		All 5 points correctly plotted to <1 small square tolerance (2) 4 points correctly plotted to <1 small square tolerance (1) 3 or less points correctly plotted to <1 small square tolerance (0) Straight line of best fit between 0 – 80 °C (1) ecf		3		3	3	3
		(ii)		As temperature increases, pressure increases (1) at a constant rate / uniformly (1) Don't accept positive correlation / proportional		2		2		2
		(iii)	I	$\frac{6}{2}$ or any equivalent ratios (1) = 3 [units] (1)		2		2	2	2
			II	94 - 3 (ecf) = 91 [units]		1		1	1	1
		(iv)		- 273 [°C]	1			1		1
	(c)			(60 - 20) or 40 (1) So $Q = 0.05 \times 720 \times (60 - 20) = 1\,440$ [J] (1)		2		2	2	
				Question 7 total	2	10	0	12	8	10